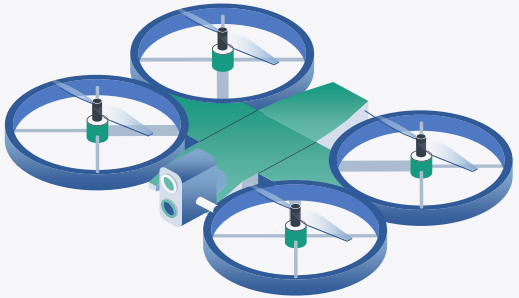
DRONE PATHFINDER AI REINFORCMENT LEARNING



By: Daniel Fridström & Hans Fredö

AGENDA

01

PROJECT CONCEPT

Short description of the project

02

GAME DESIGN

Short explination of the design of the simulation

03

[RL]

How the Reinforcement Learning work

04

RESULT

Will show the “drone” in action during Demo

05

CONCLUSIONS

Short conclusion of the work

01

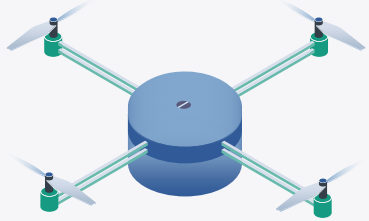
PROJECT CONCEPT



- Train an AI controlled drone to navigate a 3D environment.
- Reach checkpoints and avoid obstacles.
- Investigate how learning strategies affect success rate and navigation efficiency.

02

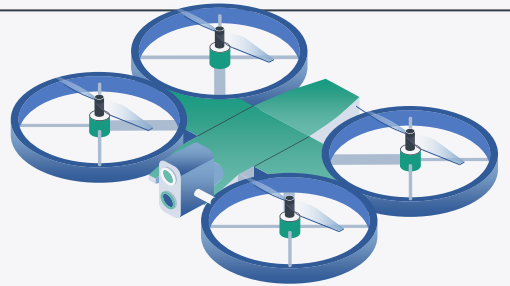
GAME DESIGN





Engine: Godot - 3D simulation.
Objective: Reach five checkpoints
Obstacles: Collisions terminates it.
Actions: Eight discrete movement actions.
Curriculum: Ten levels, increasing from 5 to 60 obstacles.
Episode: One attempt from reset until success, death or timeout.

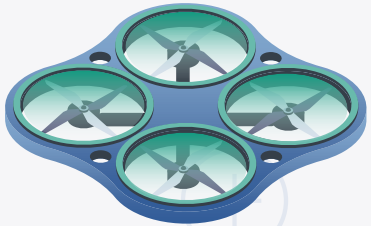
03



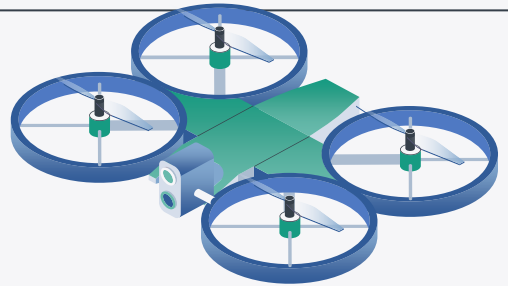
[RL] Reinforcement learning

PPO – Proximal Policy optimization.

- Placed in a simulted environment.
- It observs the current situation.
- It chooses an action.
- It receives feedback and learns from the result.

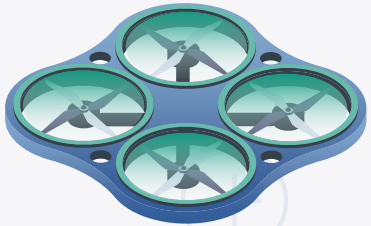


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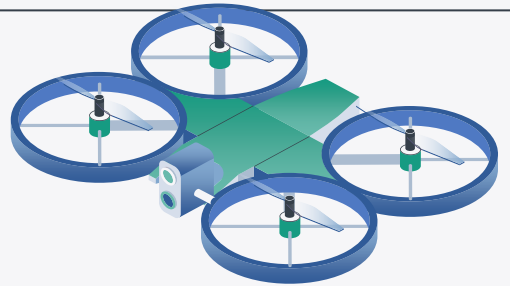


[RL] Reinforcement learning

- Positive feedback for making progress.
- Learn to reach checkpoints and complete the course.
- Better path efficiency is part of evaluating performance.
- Repeated attems help the AI discover better behaviour.

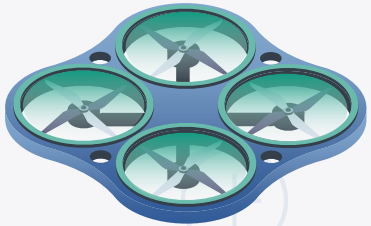


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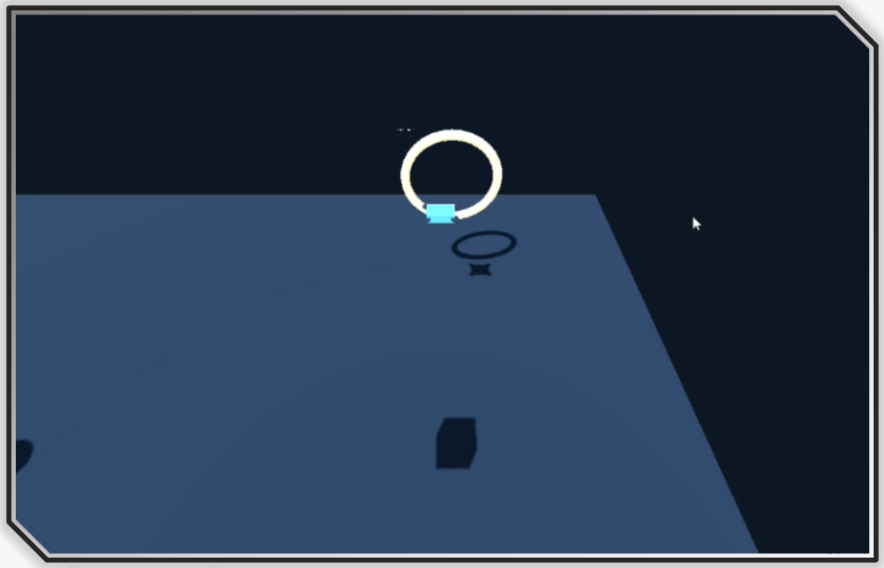


[RL] Reinforcement learning

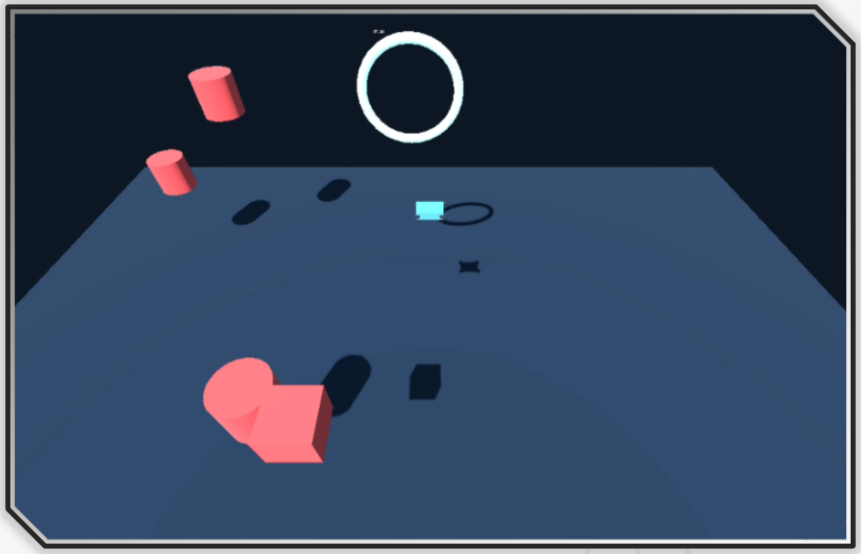
- PPO updates the drone's neural network during training.
- Training consists of repeated interaction and learning cycles.
- A curriculum gradually increases the difficulty.



04 DEMO

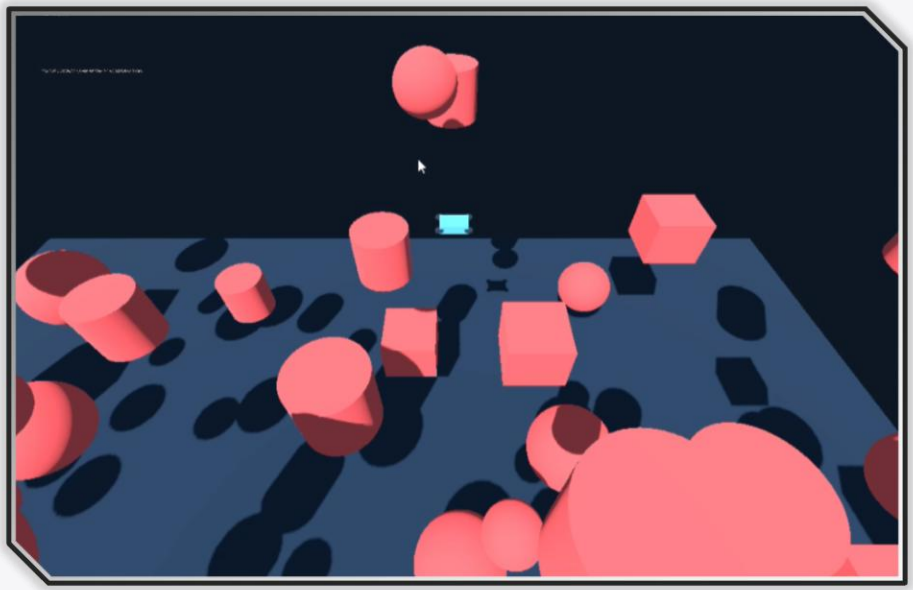


Untrained Model

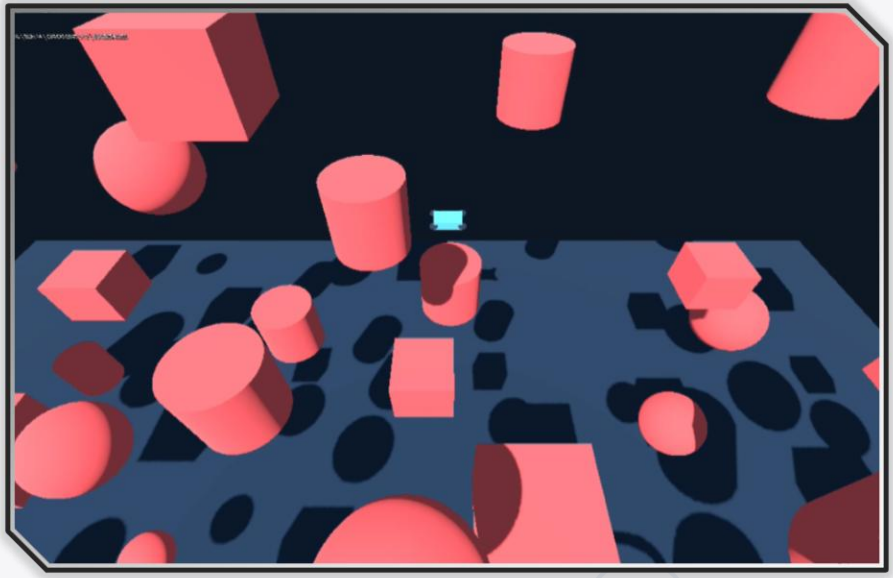


Trained Model

04 DEMO

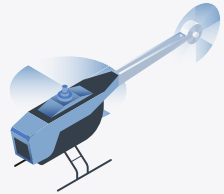


Untrained Model



Trained Model

05 CONCLUSION





Final Version